DEMAND ANALYSIS

Samir K Mahajan, M.Sc, Ph.D.,UGC-NET
Assistant Professor (Economics)

MEANING OF DEMAND

Demand is effective desire which can be fulfilled. Demand must satisfy the following pre-requisites:

- Desire for a specific commodity

- Ability to pay or sufficient resources to purchase the desired commodity
- Willingness to spend on the commodity

- Availability of the commodity at particular price, time and place.

All these mentioned requisites must be satisfied simultaneously to satisfy the meaning of demand.

DEMAND DETERMINANTS

Demand determinants refer to the factors that affect demand for commodity (a consumer good), such as:

- ❑Price of the Commodity
- ❑Income of the Consumer
- ❑Price of related goods
- ❑ Taste and preference of consumer

- ❑Growth of population
- ❑Government policy
- ❑Climatic conditions
- ❑ Income distribution
- ❑Expected change in price
- ❑Future expectation about income etc.

DEMAND DETERMINANTS

Some important determinants of demand are discussed as follows:

- **Price of the Commodity**

Normally, quantity demanded of a commodity varies inversely with its price, **ceteris paribus (i.e. other things remaining the same).** As price of a commodity rises, quantity demanded of it falls and as price of the commodity falls, quantity demanded of it rises .

- **Income of the Consumer**

Change in income of the consumer also brings about changes in demand for a commodity.
Demand for a **normal good** varies directly with income of the consumer, other things remaining the same. Normal goods are those goods whose demand increases with increase in income of the consumer and vice versa. Demand for **inferior goods** decreases with increases in income of the consumer .

DETERMINANTS OF DEMAND(contd)

❑Price of Related Goods

Goods are said to be related when they are either substitute goods or complementary goods. Substitute goods are those goods which compete with each other to satisfy a particular want. E.g. railways and airways, branded mobiles and Chinese mobile. etc . Complementary goods are those goods which are jointly demanded to satisfy a particle want. Examples of complementary goods are car and petrol, air conditioner and electricity etc.

In case of substitute goods: Quantity demanded of a commodity varies directly with the price of its substitute. *In case of complementary goods:* Quantity demanded of a commodity varies inversely with the price of its complementary goods.

❑Taste And Preferences

Demand for goods is affected by taste and preferences of the consumer which are subjective in nature, and are shaped by individual like and dislikes, faith and belief, fashions, habits, trends etc.

KINDS OF DEMAND

There are three kinds of demand relations which are usually studied under demand analysis such as: Price Demand, Income Demand and Cross Demand.

Price Demand: Price demand studies how demand for a commodity (D_x) changes with respect to change in price(P_x) , ceteris paribus (other things remaining the same).

$D_x = f(P_x)$

Income Demand: Income Demand examines how demand for commodity (D_x) changes as a result of change in income of the consumer(Y) , other things remaining the same.

$D_x = f(Y)$

Cross Demand: Cross demand studies how quantity demand of a commodity (D_x) changes as a result of change in price of its related goods (P_R), ceteris paribus . Cross demand function can be denoted as follows:

$D_x = f(P_R)$

LAW OF DEMAND

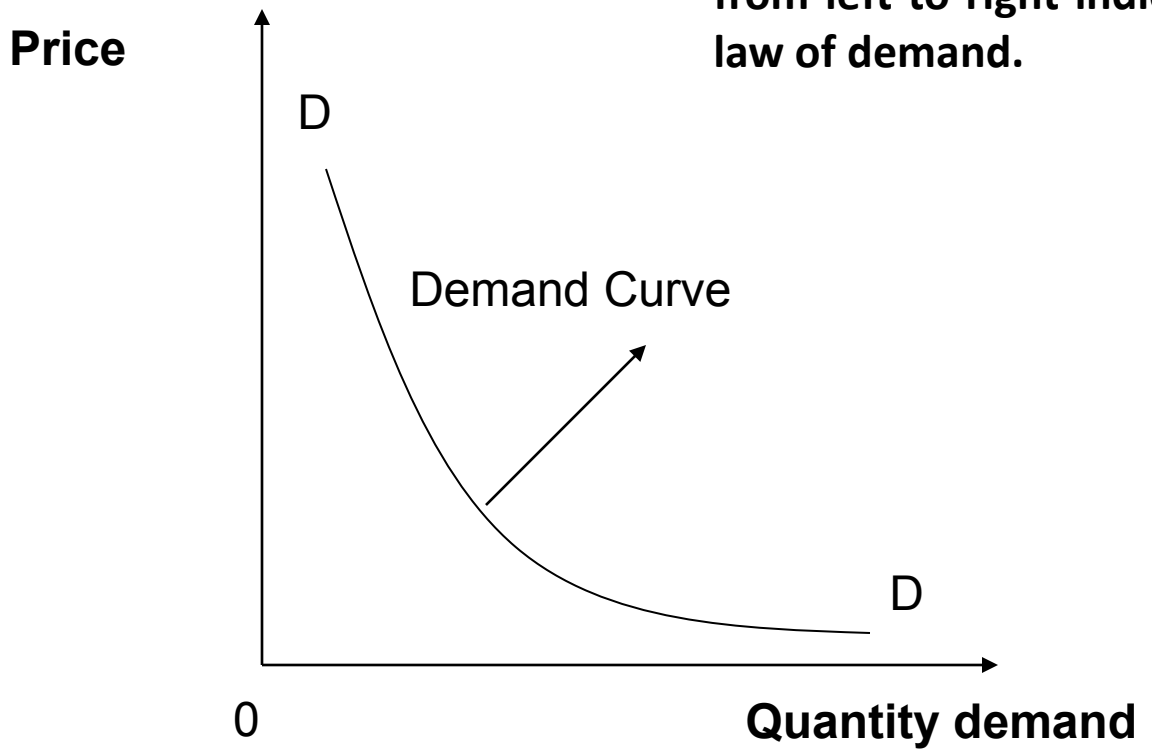
The law of demand states that normally quantity demanded of a commodity varies inversely with price, ceteris paribus.

In other words, the law of demand states that other things reaming the same, lesser quantity of a commodity will be demanded at higher prices, and more quantity of it will be demanded at lower prices.

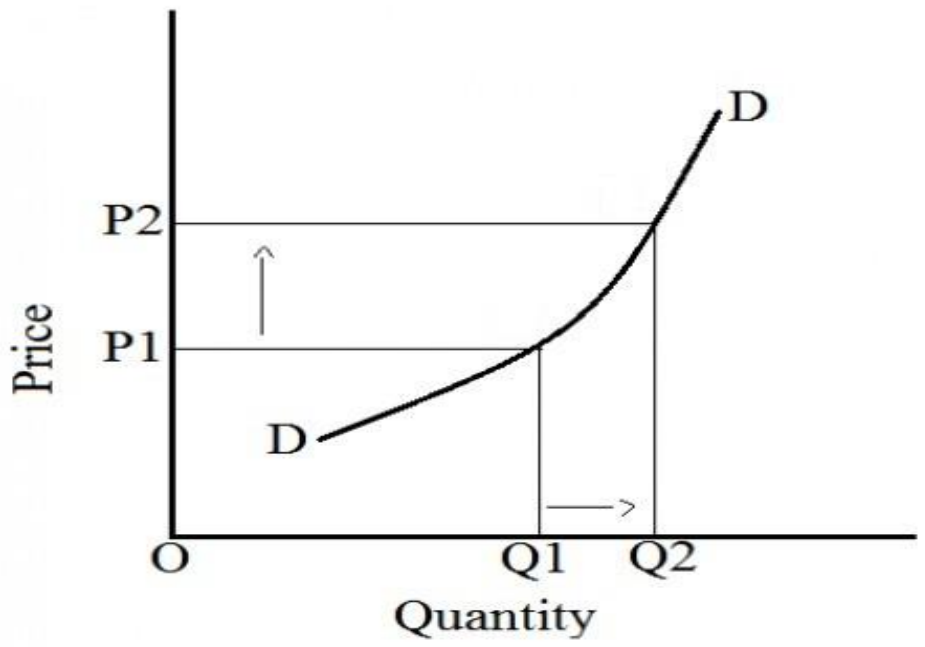
Demand curve

Demand curve is the graphical representation of the relationship between demand for a commodity (D_x) and its price (P_x) .

Normally, a demand curve slopes downward from left to right indicating the operation of the law of demand.



Exceptional Demand Curves/
Exception to Demand curve or Law of Demand



In some rare situations, the law of demand does not hold good.

In such situations, the demand curve slopes upward instead of sloping downward suggesting a rise in demand with rising price.

Cases in which this tendency is observed are referred to as exceptions to the general law of demand.

Here demand curve DD curve is known as an exceptional demand curve.

Exceptional Demand Curves contd.

Exceptions to law of demand are

- Giffen goods,
- conspicuous consumption
- conspicuous necessities,
- expected changes in price,
- extraordinary situations like natural disasters, famine, riots etc

Giffen goods:

In case of certain inferior goods called Giffen goods, when the price falls, quite often less quantity will be purchased than before because of the negative income effect and people's increasing preference for a superior commodity with the rise in their real income. Few examples of giffen goods are cheap potatoes, coarse cloth, coarse grain, etc.

Conspicuous consumption:

Some expensive commodities like diamonds, expensive cars, exorbitantly high priced mobile phones etc., are used as status symbols to display one's wealth or , to distinguish oneself from average people. The more expensive these commodities become, the higher their value as a status symbol and hence, the greater the demand for them. Law of demand does not apply here.

Conspicuous necessities:

certain things become necessities of modern life. These are purchased even if their prices rise. E.g. TV, refrigerators, mobile phones, automobiles.

Expected Changes in Price:

Expected or anticipated changes in price of a commodity in future also can affect quantity demanded of it at present. If it is expected that the price of a commodity will rise in future, the demand for it rise and vice versa.

Extraordinary situations:

War, famines, riots, natural calamities are extra ordinary situations when people's behavior becomes abnormal. Law demand does not apply in abnormal situations.

ELASTICITY OF DEMAND

Elasticity of demand is the measure of the responsiveness of quantity demanded of a commodity in response to change in a particular demand determinant (say price) while keeping other determinants constant(such as:, income, or price of related good , advertisement, growth of population and so on). Algebraically, it is defined as

$$e_D = \frac{Proportionate\ change\ in\ quantity\ demanded}{Proprtionate\ change\ in\ demdnd\ determint}$$

$$e_D = \frac{\frac{dQ}{Q}}{\frac{dZ}{Z}}$$

$$e_D = \left(\frac{dQ}{dZ}\right)\left(\frac{Z}{Q}\right)$$

Where e_D is elastic of demand
Q is quantity demanded ()
Z is any demand determinant (initial)
dQ is change in quantity demanded
dZ is change in demand determinant

CONCEPTS OF ELASTICITY OF DEMAND

There may be as many as concepts of elasticity of demand as the number of demand determinants. Most important concepts of elasticity of demand are:

- Price elasticity of demand (here the demand determinant is price of the commodity)

- Income elasticity of demand (here the demand determinant is income of consumer)

- Cross elasticity of demand (here the demand determinant is price of related goods)

PRICE ELASTICITY OF DEMAND

Price Elasticity of demand is the measure of the responsiveness of quantity demanded of a commodity in response to change in price , ceteris paribus.

Algebraically,

$$e_P = \frac{percdntage\ \ change\ in\ quanity\ demanded}{percentage\ change\ in\ price}$$

$$e_P = \frac{\frac{dQ}{Q}}{\frac{dP}{P}}$$

$$e_P = \left(\frac{dQ}{dP}\right) x (\frac{p}{Q})$$

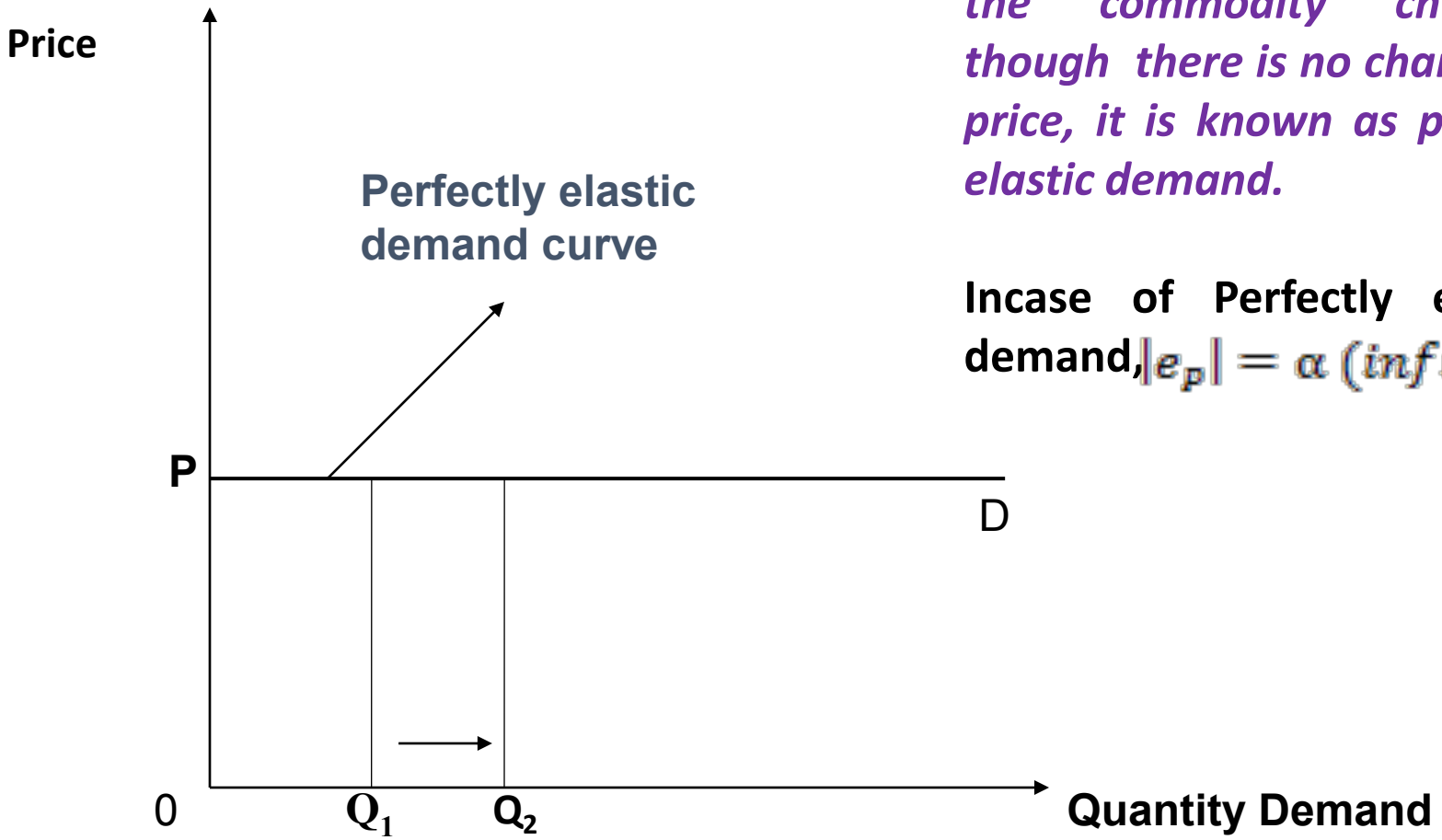
Where e_P is elastic of demand
Q is quantity demanded (initial)
P is price of the commodity (initial)
dQ is change in quantity demanded
dP change in price

***Price elasticity usually carries a negative sign because of inverse relationship between price and demand. However, it is absolute value of price elasticity of demand that determines the different degrees/kinds of price elasticity of demand.

KINDS OF PRICE ELASTICITY OF DEMAND

- **Perfectly elastic demand :** $|e_p| = \alpha\ (infinty)$
- **Elastic Demand /Relatively Elastic Demand:** $|e_P| > 1$
- **Unit Elastic Demand:** $|e_p| = 1$
- **Inelastic Demand / Relatively Inelastic Demand :** $|e_P| < 1$
- **Perfectly inelastic Demand:** $|e_p| = 0$

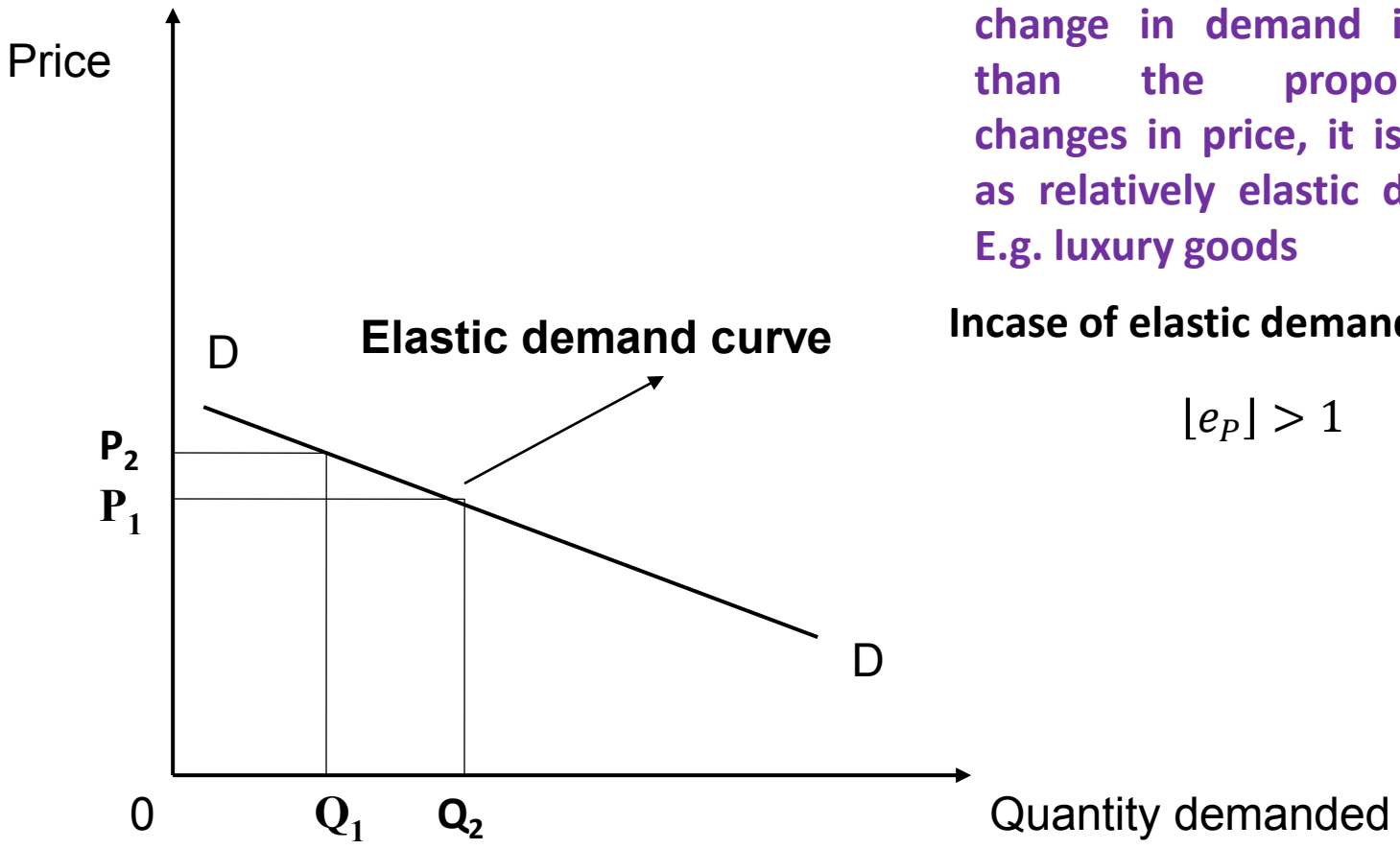
PERFECTLY ELASTIC DEMAND



When quantity demanded of the commodity changes though there is no change in price, it is known as perfect elastic demand.

Incase of Perfectly elastic demand, $|e_p| = \alpha\ (infinty)$

ELASTIC DEMAND

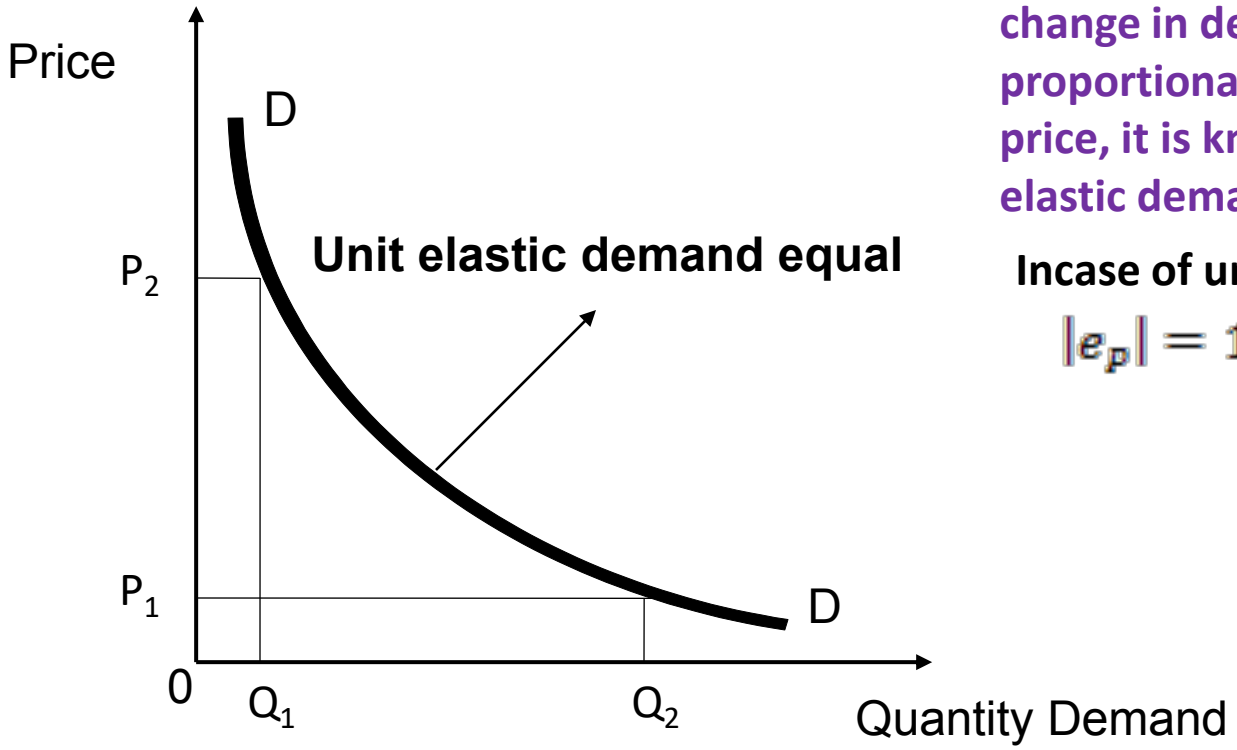


When the proportionate change in demand is more than the proportionate changes in price, it is known as relatively elastic demand. E.g. luxury goods

Incase of elastic demand,

$$|e_P| > 1$$

UNIT ELASTIC DEMAND

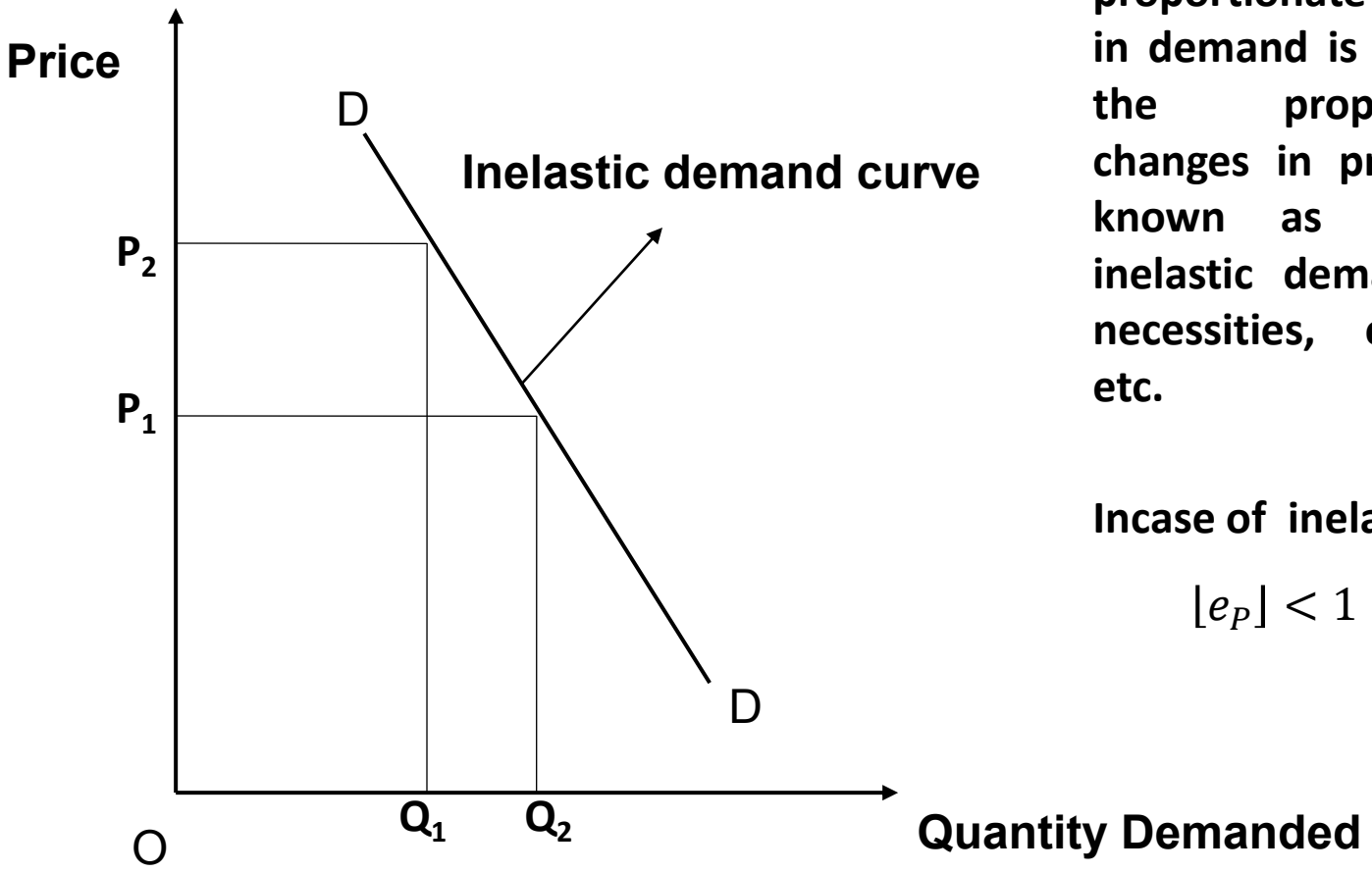


When the proportionate change in demand is equal to proportionate changes in price, it is known as unitary elastic demand.

Incase of unit elastic demand,

$$|e_p| = 1$$

INELASTIC DEMAND

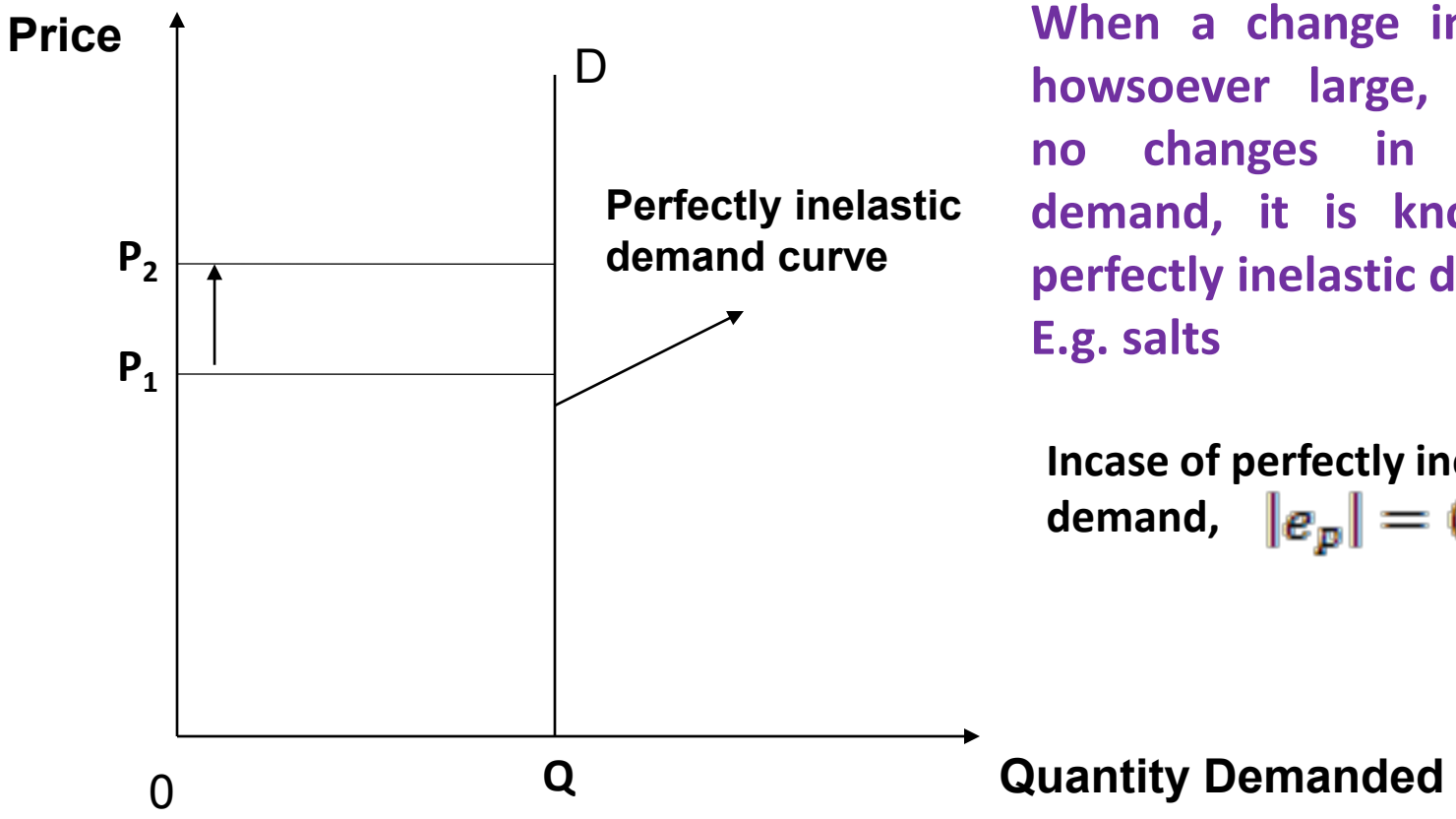


When the proportionate change in demand is less than the proportionate changes in price, it is known as relatively inelastic demand. e.g. necessities, electricity etc.

Incase of inelastic demand,

$$|e_P| < 1$$

PERFECTLY INELASTIC DEMAND



When a change in price, howsoever large, change no changes in quality demand, it is known as perfectly inelastic demand. E.g. salts

Incase of perfectly inelastic demand, $|e_p| = 0$

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Income Elasticity Of Demand

Income Elasticity of demand is the measure of the responsiveness of quantity demanded of a commodity in response to change in income of the consumer, ceteris paribus.

$$e_Y = \frac{percdntage\ \ change\ in\ quanity\ demanded}{percentage\ change\ in\ income\ of\ comusmer}$$

or, $$e_Y = \frac{dQ/Q}{dY/Y}$$

or, $$e_Y = \left(\frac{dQ}{dY}\right)\left(\frac{Y}{Q}\right)$$

Where

e_Y is income elasticity of demand
Q is the quantity demanded (initial)
Y is the income of the consumer (initial)
dQ is the change in quantity demanded
dY is the change in income

SUPPLY

Supply **indicates quantities of a commodity of a offered for sale at each possible price at a given time period, other things constant**

Determinants of Supply

- **Price of the product**
- **State of technology**
- **Prices of relevant resources**
- **Prices of alternative goods**
- **Producer expectations**
- **Number of producers/sellers in the market**

LAW OF SUPPLY

Law of supply **states that normally, the quantity supplied varies directly with its price, other things constant.**

In other words , law of supply states that lower the price, the smaller the quantity supplied and higher the price, the greater the quantity supplied.

UTILITY

Utility is defined as the power of commodity to satisfy a human want.

- **People know utility of goods by means of introspection and therefore is subjective.**
- **Being subjective, it varies from persons to persons. That is, different persons may derive different amount of utility/satisfaction from the same good.**
- **The desire for a commodity by a person depends upon the utility he expects to obtain from it.**

TOTAL UTILITY AND MARGINAL UTILITY

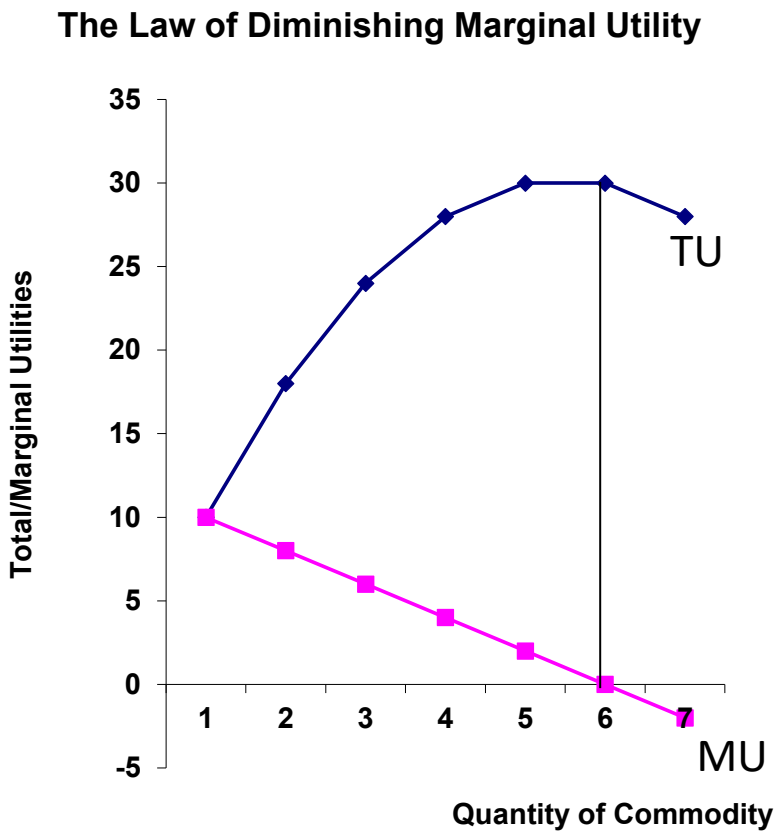
Total Utility

Total psychological satisfaction obtained by a consumer from consuming a given amount of a particular commodity is called total utility.

Marginal Utility

Marginal Utility is the extra utility derived by a consumer from the consumption of an additional unit of a particular commodity.

RELATIONSHIP BETWEEN TU AND MU



Quantity (Q)	TU (utils)	MU=dU/dQ (utils)
1	10	10
2	18	8
3	24	6
4	28	4
5	30	2
6	30	0
7	28	-2

RELATIONSHIP BETWEEN TU AND MU.

- Total utility (TU) is the sum total of marginal utilities .

TU=∑MU

- Marginal utility (MU) is the rate of change in total utility with respect to a unit change in quantity of the commodity consumed.

MU=dU/dQ
dU symbolizes change in total utility
dQ symbolizes change in quantity of commodity consumed

- When the MU decreases, TU increases at decreasing rate.
- When MU becomes zero, TU is maximum. It is a saturation point.
- When MU becomes negative, TU declines

LAWS OF CARDINAL UTILITY ANALYSIS

- Laws of Diminishing Marginal Utility
- Law of Equi-Marginal Utility

LAW OF DIMINISHING MARGINAL UTILITY

The Law of Diminishing Marginal Utility states that_the additional satisfaction derived from the additional unit of a commodity goes on diminishing.

The law highlights that while total wants of a man is unlimited, each single want is satiable. As a consumer uses more and more units of a commodity, intensity for the commodity goes on falling , and a point is reached where he does not want more of it. He is completely satisfied with the commodity which is reflected by zero marginal utility.

The law of diminishing marginal utility also serve the basis for law of law of demand or downward sloping demand curve.

CONSUMER'S EQUILIBRIUM

A consumer is in equilibrium when he maximises his utility or satisfaction by spending his given money income on different goods.

Consumer's Equilibrium in Case of Single Good

Let us take a simple model of single commodity X.

The consumer either spends his money income on the good or retains his money income.

- **In such situation, the consumer will be in equilibrium when $MU_X = P_X$**

Where, MU_X is marginal utility of commodity X

P_X is price of the commodity X.

- **If $MU_X > P_X$, the consumer can increase his well-being by purchasing more units of the commodity X.**
- **If $MU_X < P_X$, the consumer can increase his total satisfaction by cutting down the quantity of commodity X and keeping more of his income unspent.**
- **Thus, he maximises his satisfaction when $MU_X = P_X$**

Consumer's Equilibrium In case of More Than One Good and Law of Equi-Marginal Utility

The law of equi-marginal utility states that a consumer distributes his limited income among various commodities in such a way that marginal utility of money expenditure on each good is equal. This is the condition of consumer's equilibrium in case of more than one commodity.

Marginal utility of money expenditure on a good is the ratio of marginal utility of the commodity to price of it.

Consumer's Equilibrium In case of More Than One Good and Law of Equi-Marginal Utility

Thus if there are more than one commodity, the condition for equilibrium is the equality of the ratios of marginal utilities' of the individual commodities to the respective prices.

$$\frac{MU_1}{P_1} = \frac{MU_2}{P_2} = \frac{MU_3}{P_3} = \cdots \ldots \ldots \ldots \ldots .. = MU_M$$

Subject to constraint imposed by money income (Y)

$Y = P_1Q_1 + P_2Q_2 + P_3Q_3 + \ldots\ldots$

Where,

MU_1 , MU_2 , MU_3 …….. are marginal utilities of commodity 1, 2, 3,…….

P_1 , P_2 , P_3, ……. are prices of commodity 1, 2, 3,…….

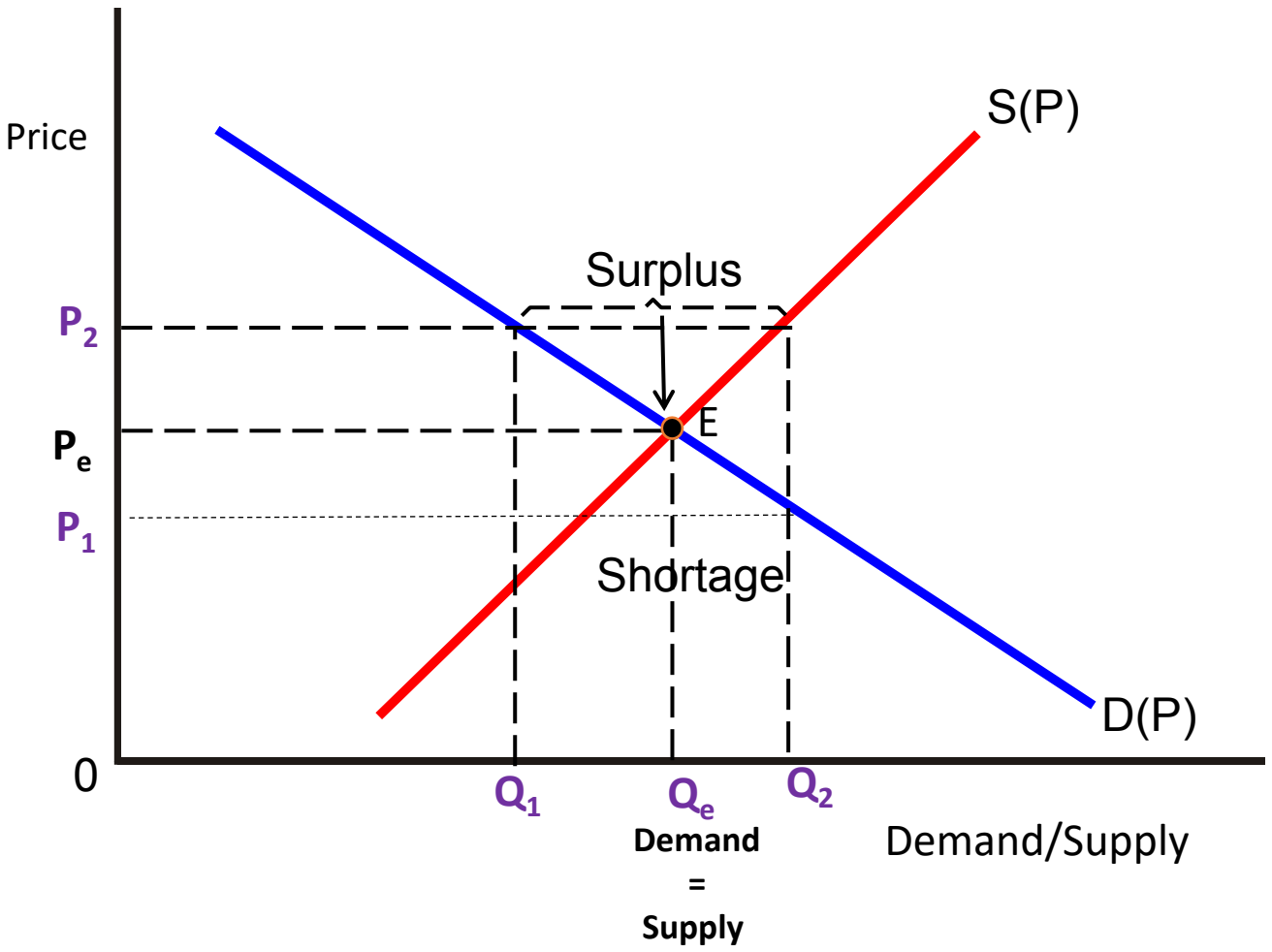
Q_1 , Q_2 , Q_3 are quantities of commodity 1, 2, 3, …….

MU_M is marginal utility of money expenditure

EQUILIBRIUM PRICE

Equilibrium price a commodity is determined at point(E) where market demand is equal to market supply.

At price P_2 , supply is more demand and thus there is surplus in the market. Price will fall causing supply to fall and demand to rise. Price will continue to fall until it reaches equilibrium price P_e at which Demand=Supply (Equilibrium point E).



At P_1, demand is more than supply and as such there is shortage in the market. Price will raise causing demand to fall and supply to rise. Price will continue to rise until it reaches equilibrium price at which Demand=Supply (Equilibrium point E).